Oussama JHIMI

Embedded Systems Engineer

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https://oussama-jhimi.github.io/portfolio

Professional Experience

Freelance Hardware/Software Engineer May 2025- Présent

- Developed a full-stack IoT system combining PCB design, embedded C++ firmware, and Python GUI with MQTT-based remote control.
- Designed and implemented a custom STM32 PCB with CAN FD support and integrated CAN transceiver for robust device control.

End of year internship in Embedded Systems February 2025 – May 2025

Marquardt Tunisie *Tunis, Tunisia*

- Designed and developed a comprehensive test box for validating embedded system blocks: DRAM, NAND Flash, microcontroller, and power management, reducing validation time by 20 % through automated testing.
- Designed a custom PCB using Altium Designer, integrating ESP32 and ADS1115 to enable accurate and automated power testing.
- Developed a modern, multi-page interface using Qt Designer and PyQt5 for managing and launching all test routines.
- Utilized Linux command-line tools for low-level testing and debugging of an ATMEL ARM Cortex microcontroller.
- Integrated a complete firmware flashing process using SAM-BA to automate microcontroller programming.

Engineering Internship in Automatic Control — Embedded Systems July 2024 – August 2024

TIM'ESS *Tunis, Tunisia*

- Performed wiring and assembly of electrical cabinets for a food coal production machine.
- Developed programmable logic controller (PLC) programs to optimize machine performance.
- Integrated STM32 F4 microcontroller to provide command and real-time control of various machine components.

Introductory Internship in Automatic Control July 2023 – August 2023

Mista Tunisia *Tunis, Tunisia*

- Performed tests and adjusted parameters, including offsets, to ensure product compliance with established quality standards.
- Collaborated with the quality department by applying the 5M and 5S principles to contribute to continuous process and product performance improvement.

Projects

Embedded System Development for IoT Control | *Altium Designer, C++, Python, PyQt5, MQTT, Mosquitto* 2025

- Developed a complete IoT solution featuring custom PCB using Altium Designer with ESP32, MH-Z19 CO sensor, and 6 relay modules.
- Created enhanced C++ sensor library supporting MH-Z19 variants with advanced calibration features.
- Developed cross-platform GUI using Python/PyQt5 for real-time monitoring and remote control via MQTT protocol.
- Implemented secure WiFi communication architecture with Mosquitto broker.

Learning Model for Local Networks Based on I2C | *STM32, I2C, Raspberry Pi ,RTOS* 2024

- Used STM32F4 as the I^2C master, with Raspberry Pi, AHT10 sensor, and an LCD display with I^2C module as slaves to demonstrate the operation of the I^2C bus.
- Implemented temperature measurement using the AHT10 sensor and real-time data display on an LCD screen (I^2C module), stored data in InfluxDB and visualized it through Grafana on a Raspberry Pi server.

Embedded Intelligence in an IoT Network | *STM32, ESP32, UART, Blynk* 2024

- Transfer of temperature values measured by STM32 to ESP32 using UART to ensure communication between the two boards. Sending the values to Blynk using the Wi-Fi network.

Building a Custom Operating System for Raspberry Pi 3 | *Embedded Linux, ARM architecture, U-Boot, Kernel* 2024

- Set up a cross-compiler, configured U-Boot bootloader, and compiled the Linux kernel.
- Created a minimal RootFS and implemented the complete boot process.

Plant Disease Detection | *Raspberry Pi, Deep Learning, YOLOv7, Google Colab, Roboflow, Kaggle* 2024

- Designed and developed an advanced plant disease detection system using the YOLOv7 model.
- Trained the model on Google Colab with an annotated dataset from Roboflow and integrated it with a Raspberry Pi serving as a camera server.

Light Dimmer | *Arduino, PCB* 2023

- Designed and developed a light dimming system with Arduino and other components to provide precise and adjustable lighting control.

Skills

Embedded Systems: STM32, Raspberry Pi, ATMEL ARM, ESP32,Arduino.
Communication protocol:I2C, UART, CAN, SPI, MQTT, Modbus Protocol, LoRa, RS485.
Programming Languages : C, C++, Python, Java, VHDL, Assembly.
Tools: VS Code, GIT, Altium Designer, Proteus ISIS, LABView, TIA portal, LTspice, RTOS, STM32cubeIDE, Qt Designer, Yocto.
Soft Skills: Teamwork, Problem-Solving, Time Management, Effective Communication.
Languages: English(B2), French(B2), Arabic (native language), Italian (basic knowledge).

Education

- Engineering Cycle in Electrical Automation - National Engineering School of Gabès 2022-2025
- Preparatory Cycle MP - ISSAT Gabes 2018-2021
- Baccalaureate, Mathematics - Abou Kacem Chebbi High School Tozeur 2018

- Altium Designer – MAT
- Embedded linux – coursera
- c,c++ – coursera

Organization

Alliance Of Engineers Club (AOE): *Project Manager - Robotics Lead* 2023-2024

National Robotics Event ENIGROBOTS 5.0: *Head of the All-Terrain Committee*

International Robotics Event ENIGROBOTS 4.0: *Member of the Death Ring Committee*